# Julien HORGUEDEBAT

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## Presentation

Student in the Master program in Advanced Mathematics at the Université Marie et Louis Pasteur in Besançon (France)

## Education

**Université Marie et Louis Pasteur** *Sept 2023 – present*
*Master degree in Advanced Mathematics*

- **Coursework:** Numerical Analysis, Partial and Ordinary Differential Equations, Student Seminar, Fields Theory, Algebra, Complex Analysis, Fourier Analysis, Topology, Probabilities

**Université Marie et Louis Pasteur** *Sept 2022 – June 2023*
*Bachelor degree in Fundamental Mathematics*

- Result: 15.415/20
- **Coursework:** Algebra, Metric spaces, Integration Theory, Functional Analysis, Probabilities, Affine Structure, Numerical Analysis, Differential calculus

**Université Marie et Louis Pasteur** *Sept 2022 – present*
*Magistère in Mathematics*

- **Coursework:** Epistemology, Statistics, Functional Analysis, Introduction to C, Colloquium

## Experience

**Research Internship** *Besançon, FR*
*Laboratoire de Mathématiques de Besançon - UMR 6623* *May 2024 – Jul 2024*

- Implementation of numerical methods to solve integral equations using the Nyström method.

**Handler** *Besançon, FR*
*Grand Besançon Métropole* *June 2023*

## Seminars Attended

**Laboratoire de Mathématiques de Besançon - UMR 6623** *Sept 2025 – present*
*Numerical Analysis and Scientific Computing*

- Weekly attending in the workgroup and seminar of the Numerical Analysis and Scientific Computing team of the laboratory.

**Research Initiation School of the LmB** *Jan 2023*
*Winter School of the Maths Master of LmB* *Jan 2025*
*Laboratoire de Mathématiques de Besançon - UMR 6623* *Dec 2025*

- Conferences by members of the laboratory of mathematics

**Lectures Sophie Kowalevski** *4-6 June 2024*
*Centre Henri Lebesgue - Angers*

- Short-courses in algebra and analysis/probabilities.

## Academic Projects

**Moment Problems** *Oct 2025 – present*
*Université Marie et Louis Pasteur; directed by Thierry DAUDE and Bastien POLLIZI*

- Methods of resolution for moment problems and numerical implementation
- Tools: Julia, LaTeX

**Master 1 thesis: Fourier Transform on a Finite Commutative Group** *Jan 2024 – May 2024*
*Université Marie et Louis Pasteur; directed by Cécile ARMANA; group of 2*

- Project about group theory, Fourier transform on finite commutative groups, Fast Fourier Transform and applications to polynomial multiplication.
- Tools: Python, LaTeX

**Bachelor Project: On the existence of invariant measures under the action of a group and the Banach-Tarski paradox** *Jan 2023 – May 2023*
*Université Marie et Louis Pasteur; directed by Marsault CHABAT; group of 3*

- Project about group theory, the Banach-Tarski paradox and the existence of non-measurable sets in $\mathbb{R}$.
- Tools: LaTeX

## Skills

**Programming languages:** Python (Advanced), Julia (Advanced), LaTeX(Advanced), C (basic)
**Languages:** French (Mother Tongue), English (Technical), Spanish (Basic)
**First aid certification (PSC1)**